Submission

AUTHORS:

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TITLE:

Healthcare Analytics Challenges: A Three-Pillar Framework Connecting Analytics Maturity, Workforce Agility, and Technical Enablement

ORIGINAL FILE

96541-1413317-2-SM.docx 2026-03-30

SUPP. FILES

Add a Supplementary File

SUPPLEMENTARY AND FIGURE/APPENDIX FILES

Figures

File	Date	Action	Thumbnail
1. **Figure 1**. Human-in-the-Loop Knowledge Governance (HITL-KG) Architecture. A Clinical User submits a natural language query (step 1) to a Conversational AI with NLP Engine and SQL Generation, which generates SQL and queries the Healthcare Data Warehouse (steps 2-3). The system then presents its logic to	2026-03-30	Edit | Delete Change to Figure Number: 1 Update	

Support

confirms and delivers insights back to the user (steps 5-6) or loops back for correction. On confirmation, the validated query triple is stored in Organizational Memory (step 7, dashed line), which informs future queries alongside a Knowledge Base of ontologies and best practices. architecture.mmd.png

2.

Figure 2. Six-step Validated Query Cycle for Human-in-the-Loop Knowledge Governance. Step 1 (Query): analyst asks a natural language question. Step 2 (Generation): system generates candidate SQL. Step 3 (Validation): AI explains SQL logic and results; analyst confirms intent. If incorrect, the cycle loops back to Generation. If correct, Step 4 (Storage): validated triple is stored in organizational memory. Step 5 (Retrieval): future queries match against validated triples. Step 6 (Persistence):

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2

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validated queries. Dashed arrows connect the cycle to three outcome pillars: Storage advances Analytics Maturity, Persistence stabilizes Workforce Agility, and Retrieval increases Technical Enablement. knowledge-cycle.mmd.png

Multimedia Appendices

File	Date	Action	Thumbnail
1. **Multimedia Appendix 1.** Validated Query Triple examples illustrating the Human-in-the-Loop Knowledge Governance (HITL-KG) knowledge capture artifact. Each triple comprises a natural language intent, executable SQL, and rationale metadata for three clinical analytics scenarios: diabetes care monitoring, heart failure readmission cohort identification, and quality metric aggregation. multimedia_appendix_1.docx	2026-03-30	Edit | Delete Change to Appendix Number: 1 Update	

TOC/Feature image for homepage

File	Date	Action	Thumbnail
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SUBMITTER

Samuel Harrold [Change to another user]

DATE SUBMITTED

March 30, 2026 - 12:37 AM

SECTION

Advanced Data Analytics in eHealth

EDITOR

None assigned

AUTHOR COMMENTS

Dear Editor,

I am writing to resubmit the manuscript "Healthcare Analytics Challenges: A Three-Pillar Framework Connecting Analytics Maturity, Workforce Agility, and Technical Enablement" (ms#91493) for consideration as a *Viewpoint* article in JMIR Medical Informatics.

Response to Decision E2

I appreciate the editor's feedback that the work is "interesting" and the invitation to resubmit after addressing the length concern and submission type.

Summary of Revisions

The manuscript has been revised to meet Viewpoint article requirements:

Structural changes:

- Reduced to approximately 4,600 body words (within the 5,000-word limit)
- Converted from IMRD structure (Introduction, Methods, Results, Discussion) to descriptive thematic headers appropriate for Viewpoint articles
- Converted the structured abstract to an unstructured flowing narrative (302 words)

Content focus:

The revised manuscript retains the core intellectual contributions while removing the exhaustive methodology and literature review sections:

- *Human-in-the-Loop Knowledge Governance (HITL-KG) framework:* The socio-technical

framework, resolved through Lean "Standard Work"
- *Three-Pillar Assessment Rubric:* A structured self-assessment tool with evidence-based indicators across analytics maturity, workforce agility, and technical enablement
- *Three pillars of evidence:* Synthesized evidence on analytics maturity, workforce agility, and technical enablement

Quality assurance:

- All citations updated to 2024-2026 data where available
- Framework advances a prescriptive Viewpoint position grounded in descriptive evidence; conversational AI framed as enabler, not solution
- Competing interests disclosed per JMIR requirements

Article Type

This submission is a *Viewpoint* article: an evidence-based scholarly perspective on mitigating institutional amnesia in healthcare analytics, grounded in peer-reviewed literature and knowledge management theory.

Thank you for your continued consideration.

Respectfully,

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Status

Unassigned

STATUS LAST MODIFIED

2026-03-30

Submission Metadata

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***PRINCIPAL ADDRESS FOR EDITORIAL CORRESPONDENCE.**

Yes

Title and Abstract

Title

Healthcare Analytics Challenges: A Three-Pillar Framework Connecting Analytics Maturity, Workforce Agility, and Technical Enablement

Abstract

Healthcare organizations face a "Triple Threat" of low analytics maturity, high workforce instability, and semantic technical barriers that together produce a crisis of "Institutional Amnesia." High leadership turnover, persistent digital health

SECI model of knowledge creation, the root cause is a "Socialization Failure": high turnover fractures the social networks required for mentorship, rendering the traditional apprenticeship model of informatics unsustainable. To address this failure, we employ a Design Science Research (DSR) approach, synthesizing evidence from healthcare informatics, knowledge management, and natural language processing (2024-2026 workforce and NL2SQL literature) to develop a socio-technical framework called Human-in-the-Loop Knowledge Governance (HITL-KG). HITL-KG is designed to shift the locus of organizational knowledge from volatile human memory to durable semantic artifacts called "Validated Query Triples," each comprising a natural language intent, executable SQL, and rationale metadata. By embedding knowledge capture into the daily workflow of query generation, the framework aims to convert the ephemeral act of analytics into permanent institutional assets. The accompanying Three-Pillar Assessment Rubric provides a structured self-assessment tool that enables organizations to identify compounding vulnerabilities across analytics maturity, workforce agility, and technical enablement. A critical objection, the "Validator Paradox" (who validates the AI when experts leave?), is resolved by reframing validation through Lean "Standard Work": each validated query establishes the current known standard rather than eternal truth, functioning as a "knowledge ratchet" that prevents regression. By decoupling analytical capability from individual tenure, healthcare systems can ensure that analytics maturity advances even as the workforce evolves. This paper proposes and theoretically motivates the framework; empirical validation is deferred to a companion study.

Journal Transfer Preferences

Journal Name

Author Preferences

Primary Section

Indexing

Keywords

institutional amnesia ; medical informatics ; socio-technical systems ; query governance ; natural language processing ; knowledge management ; personnel

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